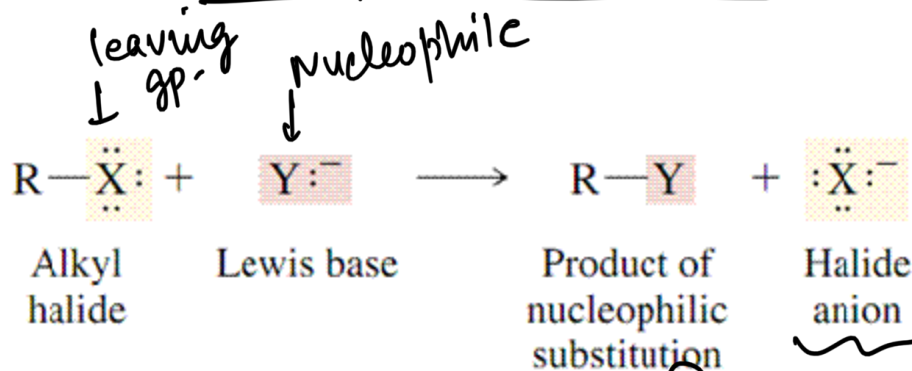
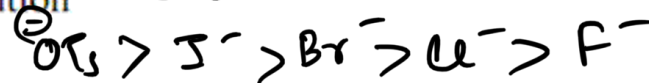


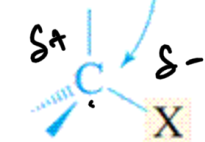
Nucleophilic Substitution reactions



1935
Hughes & Ingold
Aliphatic

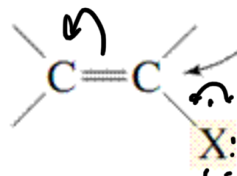


sp^3 -hybridized carbon

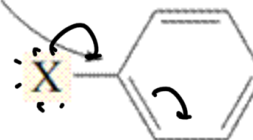


Alkyl halide

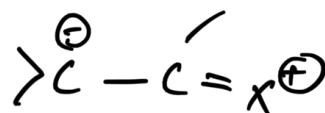
sp^2 -hybridized carbon

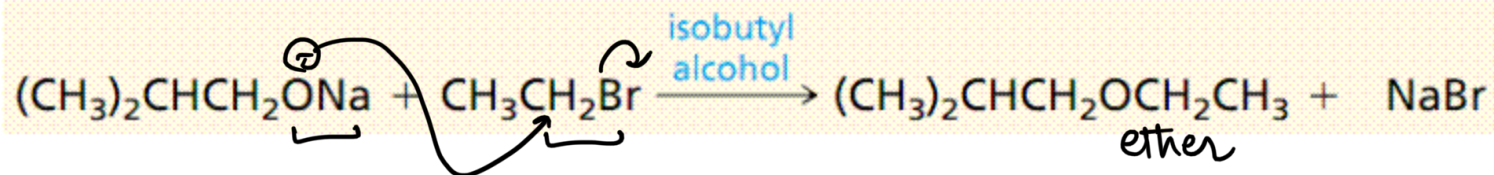


Alkenyl halide

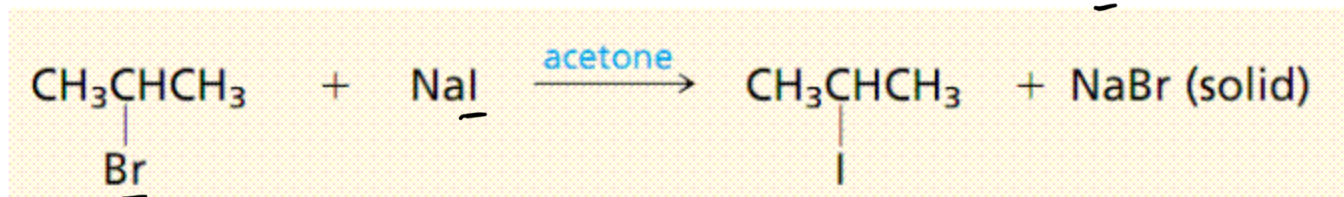
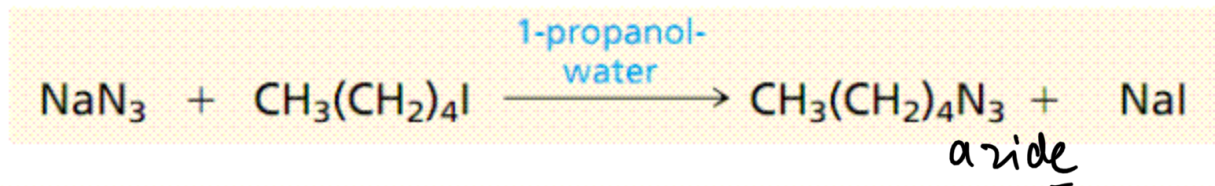
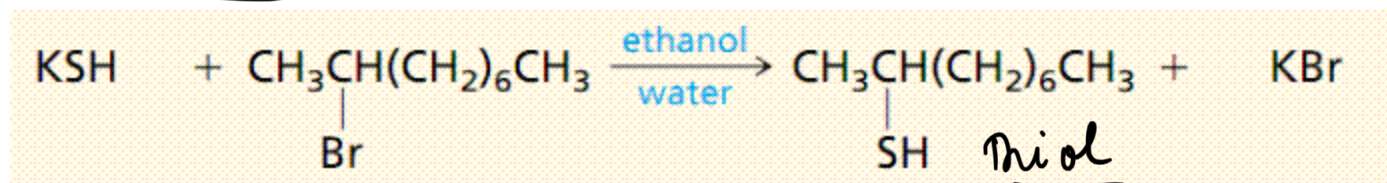
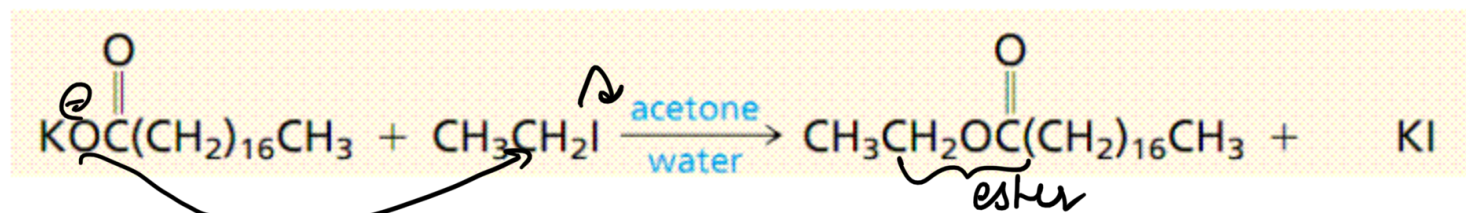


Aryl halide





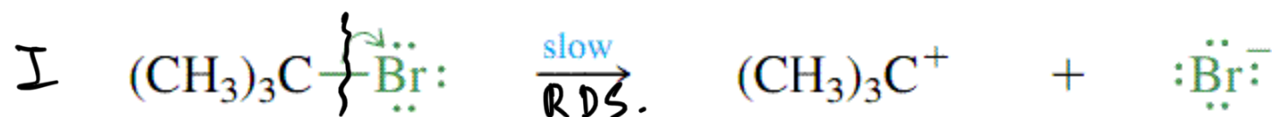
Fggs



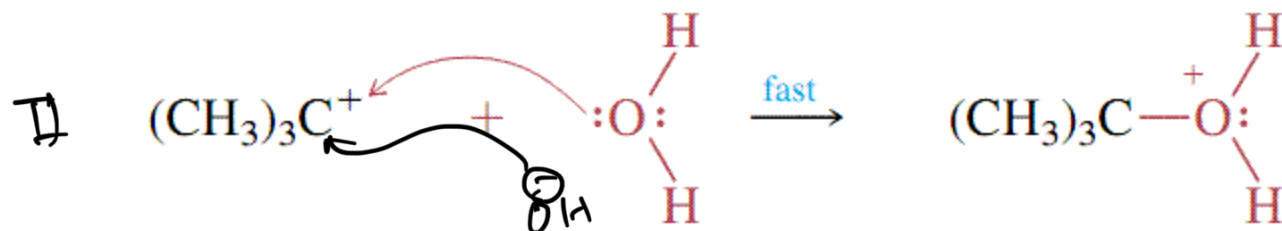


$$\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]$$

S_N1
Mechanism



Homolytic cleavage of C-X

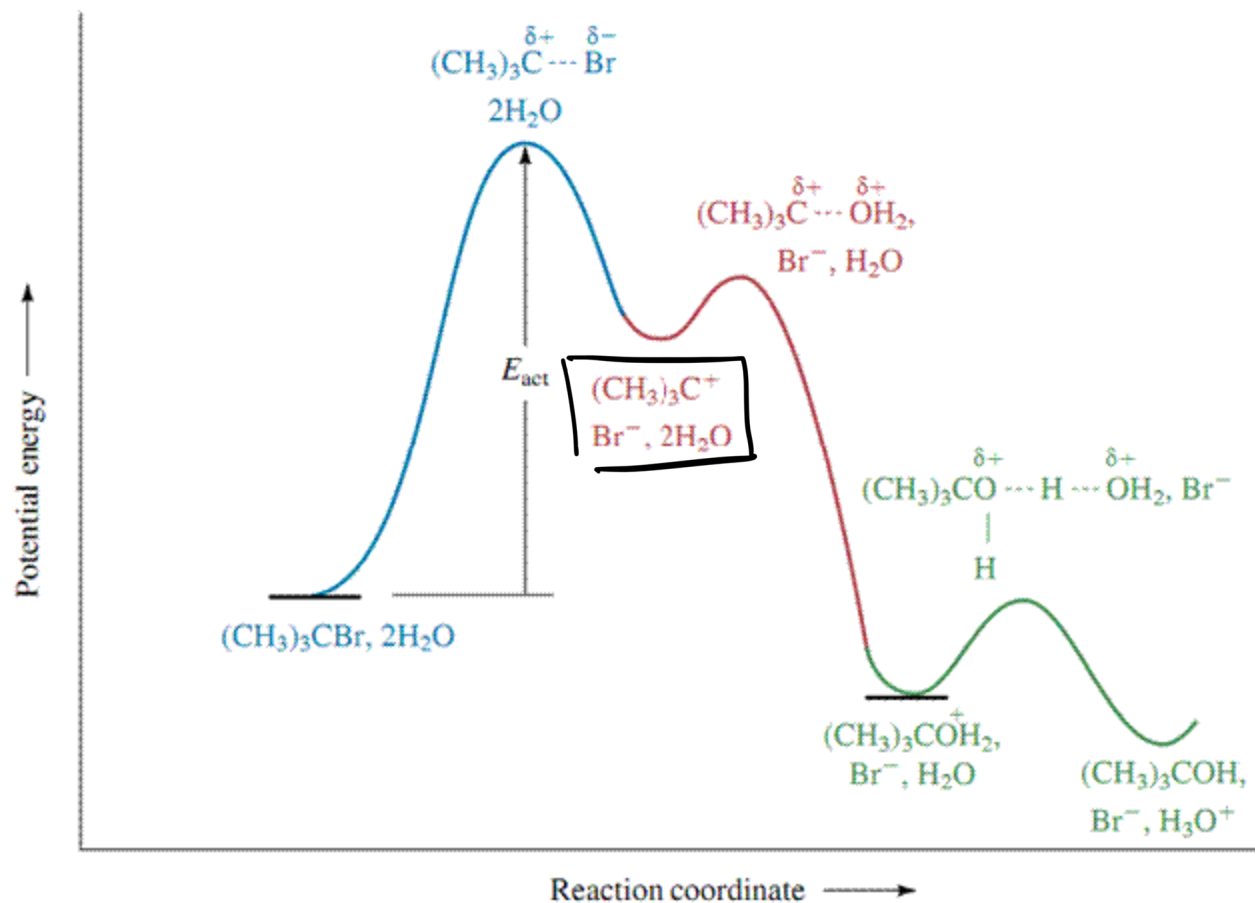


Attack of Nu



loss of ~~H⁺~~

Energetics of S_N1



order of reactivity of halides in S_N1

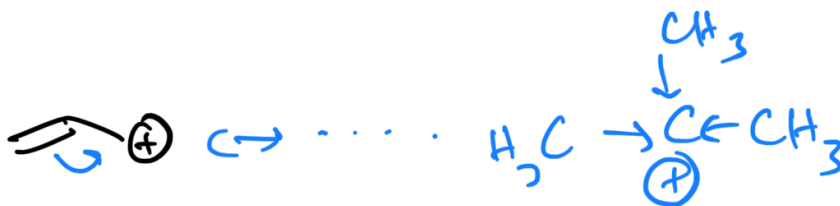
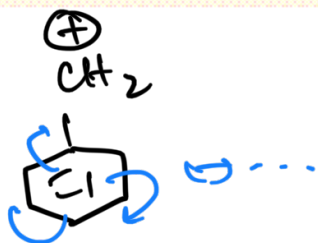
benzyl, allyl > tertiary > sec > primary > methyl halide

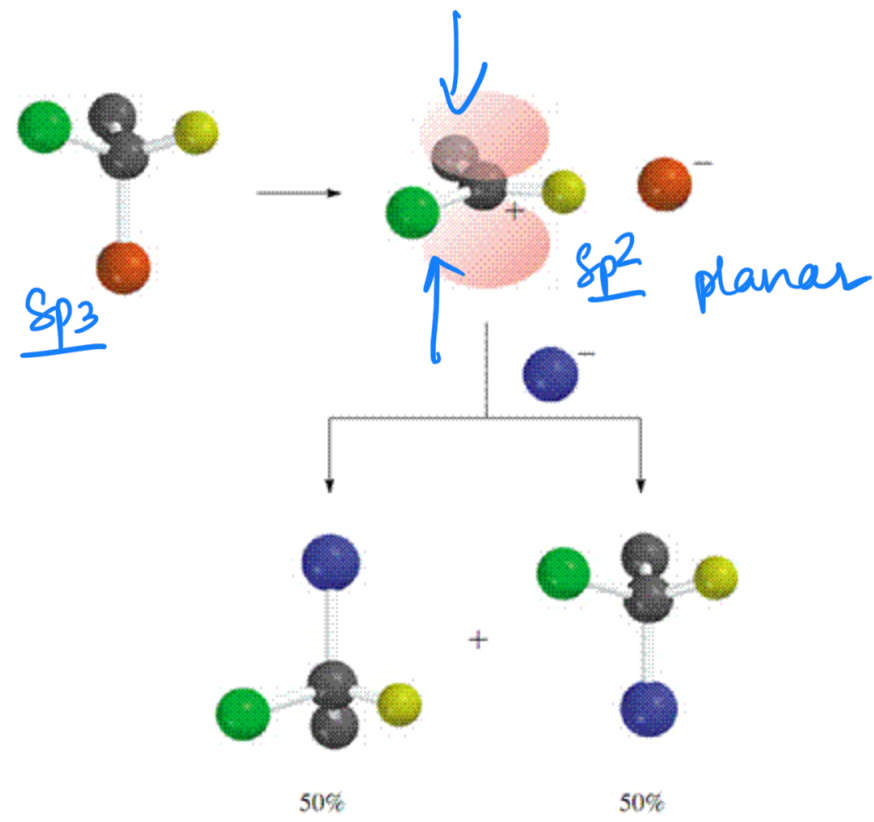
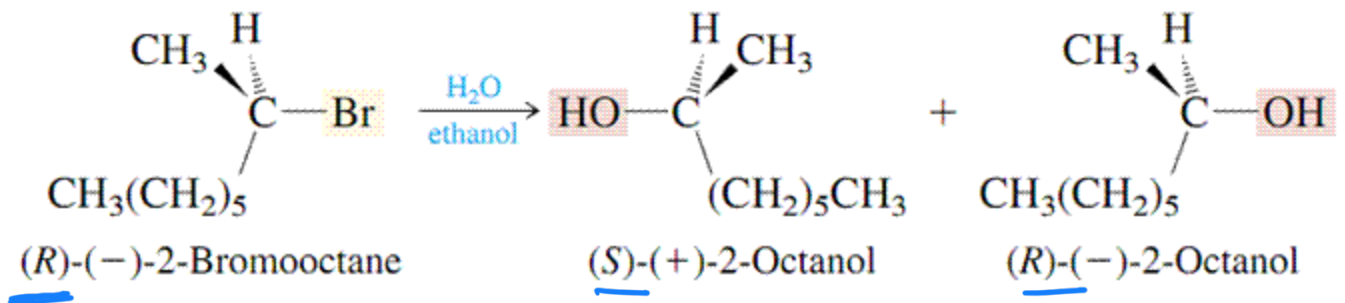
Reactivity of Some Alkyl Bromides Toward Substitution by the S_N1 Mechanism*

Alkyl bromide	Structure	Class	Relative rate [†]
Methyl bromide	CH_3Br	Unsubstituted	1
Ethyl bromide	$\text{CH}_3\text{CH}_2\text{Br}$	Primary	2
Isopropyl bromide	$(\text{CH}_3)_2\text{CHBr}$	Secondary	43
<i>tert</i> -Butyl bromide	$(\text{CH}_3)_3\text{CBr}$	Tertiary	100,000,000 →

*Solvolysis in aqueous formic acid.

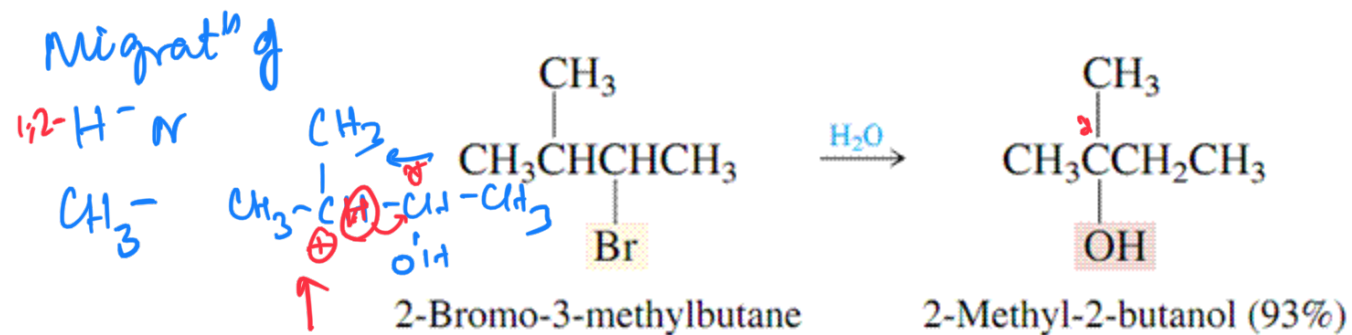
[†]Ratio of rate constant k for indicated alkyl bromide to k for methyl bromide at 25°C.



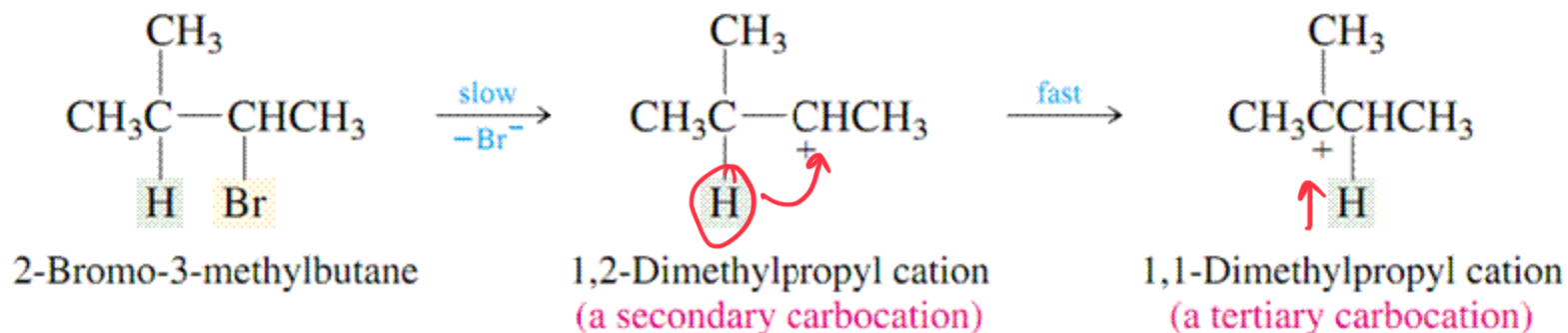


Stereochemistry of
 $\text{S}_{\text{N}}1$

Racemic mix
in general



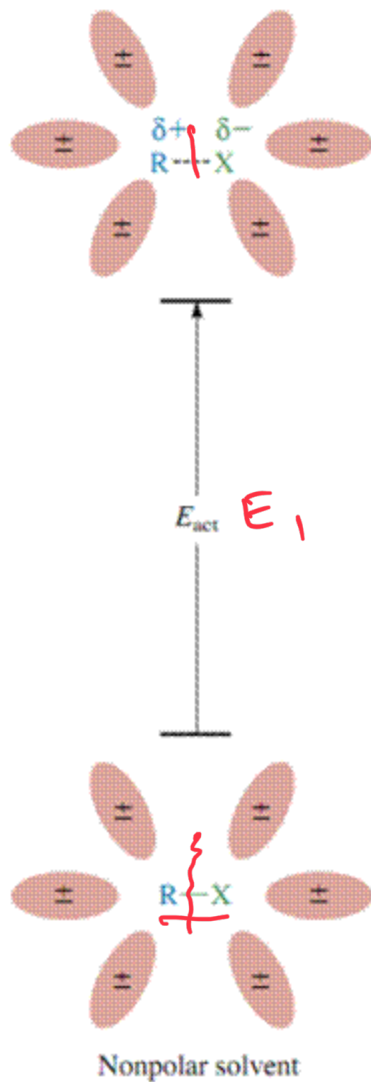
Rearrangement of Carbocation in $\text{S}_\text{N}1$: less stable
↓
more stable C^+



Relative Rate of $\text{S}_\text{N}1$ Solvolysis of *tert*-Butyl Chloride as a Function of Solvent Polarity* Polar Protic Solvent

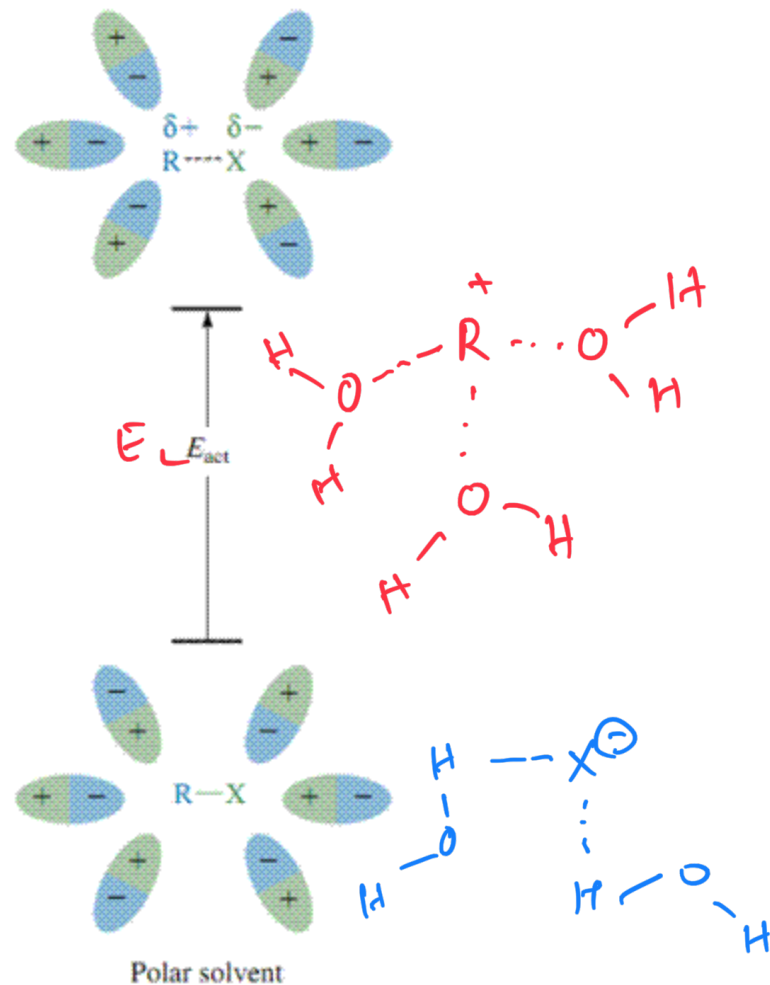
Solvent	Dielectric constant ϵ	Relative rate
Acetic acid	6	1
Methanol	33	4
Formic acid	58	5,000
Water	78	150,000

*Ratio of first-order rate constant for solvolysis in indicated solvent to that for solvolysis in acetic acid at 25°C.



Transition state is more polar than starting state; polar solvent can cluster about transition state so as to reduce electrostatic energy associated with separation of opposite charges.

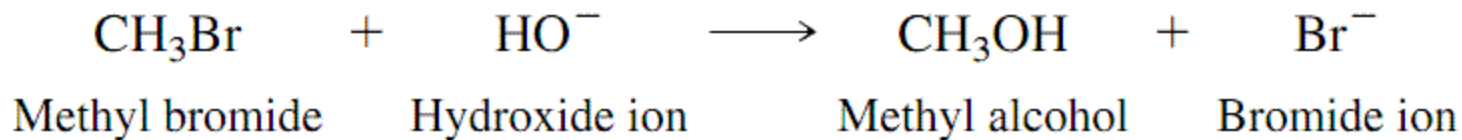
$$E_1 > E_2$$



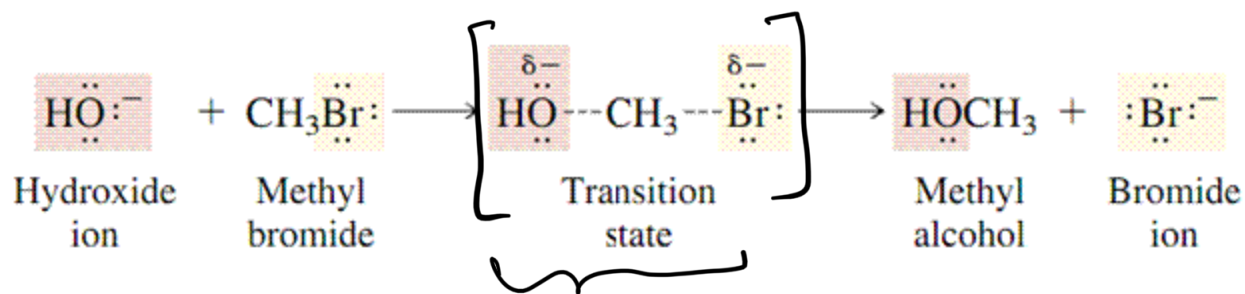
SN² → bimolecular

RX

NU → RDS .

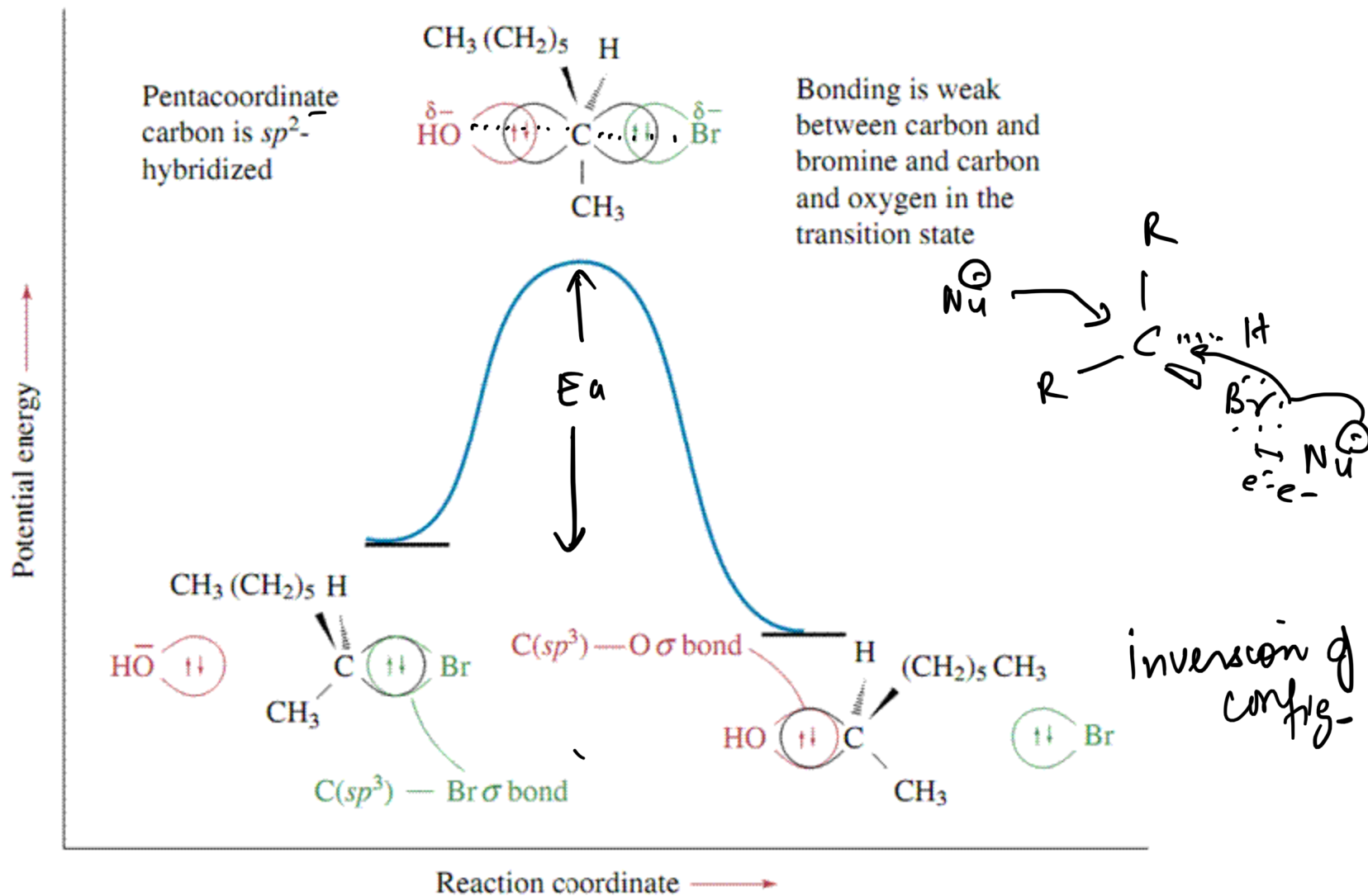


$$\text{Rate} = k[\text{CH}_3\text{Br}][\text{HO}^-] \quad \text{SN}^2$$



Concerted
one-step

Energy profile for S_N2

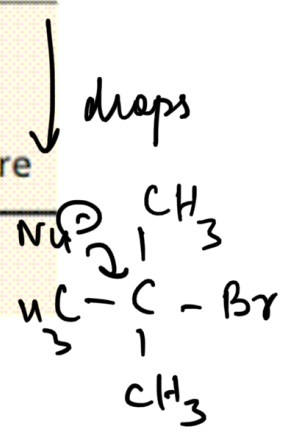


Reactivity of Some Alkyl Bromides Toward Substitution by the S_N2 Mechanism*

Alkyl bromide	Structure	Class	Relative rate [†]
Methyl bromide	→ CH ₃ Br	Unsubstituted	221,000
Ethyl bromide	CH ₃ CH ₂ Br	Primary	1,350
Isopropyl bromide	(CH ₃) ₂ CHBr	Secondary	1
<i>tert</i> -Butyl bromide	(CH ₃) ₃ CBr	Tertiary	Too small to measure

*Substitution of bromide by lithium iodide in acetone.

[†]Ratio of second-order rate constant *k* for indicated alkyl bromide to *k* for isopropyl bromide at 25°C.



Increasing rate of substitution
by the S_N2 mechanism



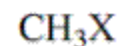
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Tertiary

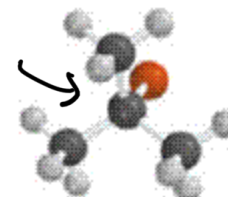
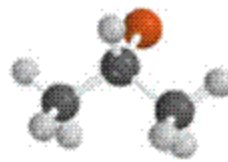
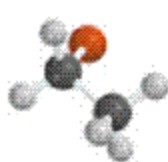
Least reactive,
most crowded

Secondary

Primary

Methyl

Most reactive,
least crowded



Effect of Chain Branching on Reactivity of Primary Alkyl Bromides Toward Substitution Under S_N2 Conditions*

Alkyl bromide	Structure	Relative rate [†]
Ethyl bromide	$\text{CH}_3\text{CH}_2\text{Br}$	1.0
Propyl bromide	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	0.8
Isobutyl bromide	$(\text{CH}_3)_2\text{CHCH}_2\text{Br}$	0.036
Neopentyl bromide	$(\text{CH}_3)_3\text{CCH}_2\text{Br}$	0.00002

*Substitution of bromide by lithium iodide in acetone.

[†]Ratio of second-order rate constant k for indicated alkyl bromide to k for ethyl bromide at 25°C.

1° R^x



Nucleophilicity of Some Common Nucleophiles

Reactivity class	Nucleophile	Relative reactivity*
Very good nucleophiles	I^- , HS^- , RS^-	$>10^5$
Good nucleophiles	Br^- , HO^- , RO^- , CN^- , N_3^-	10^4
Fair nucleophiles	NH_3 , Cl^- , F^- , RCO_2^-	10^3
Weak nucleophiles	<u>H_2O, ROH</u>	1
Very weak nucleophiles	RCO_2H	10^{-2}

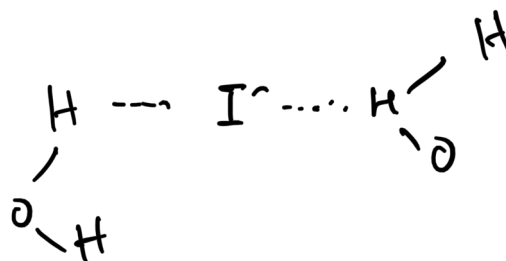
*Relative reactivity is $k(\text{nucleophile})/k(\text{methanol})$ for typical $\text{S}_{\text{N}}2$ reactions and is approximate. Data pertain to methanol as the solvent.

Relative Rate of S_N2 Displacement of 1-Bromobutane by Azide in Various Solvents*

Solvent	Structural formula	Dielectric constant ϵ	Type of solvent	Relative rate
Methanol	CH ₃ OH	32.6	Polar protic	1
Water	H ₂ O	78.5	Polar protic	7
Dimethyl sulfoxide	(CH ₃) ₂ S=O	48.9	Polar aprotic	1300
N,N-Dimethylformamide	(CH ₃) ₂ NCH=O	36.7	Polar aprotic	2800
Acetonitrile	CH ₃ C≡N <i>polar aprotic</i>	37.5	Polar aprotic	5000 →

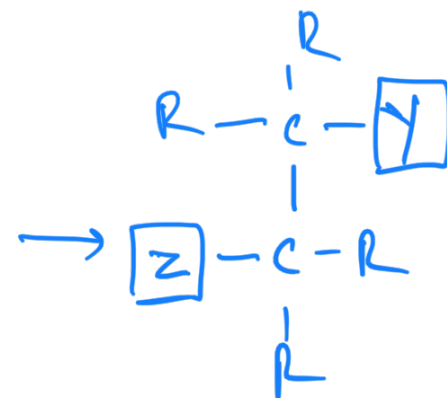
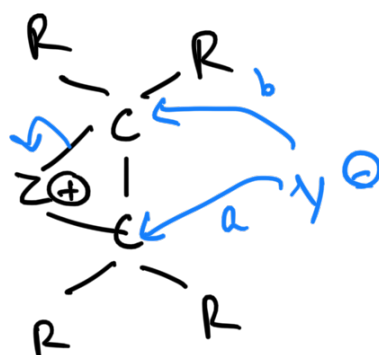
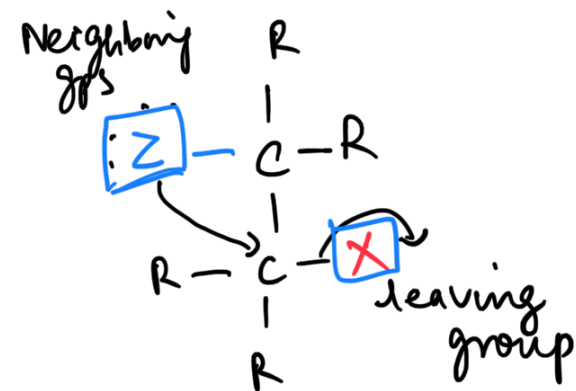
*Ratio of second-order rate constant for substitution in indicated solvent to that for substitution in methanol at 25°C.

<http://www.youtube.com/watch?v=TnY1S5IdVqI>

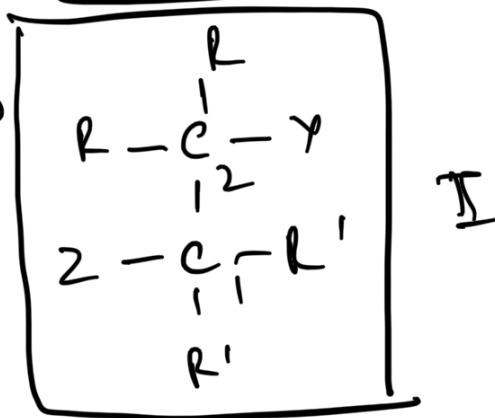
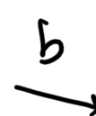
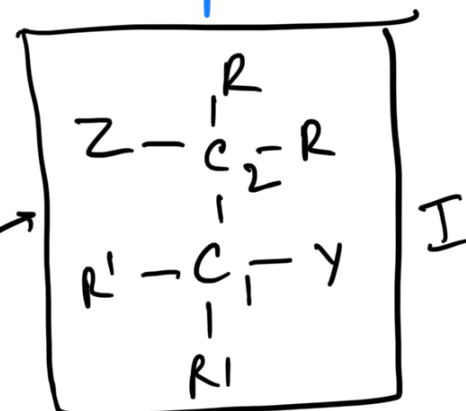
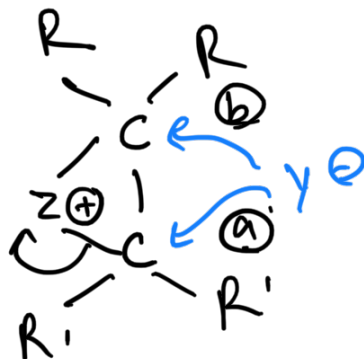
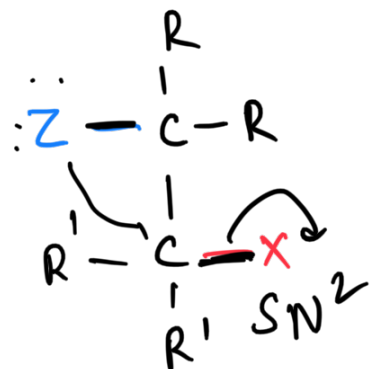


Neighbouring Group Participation (NGP)

- Anchimeric assistance
- Takes place in S_N reactions
- Rate of reaction is exceptionally high (at least 50 times)
- Reaction proceeds with 100% retention of configuration.
- | | |
|---------------|------------------------------------|
| Syn - syn |] Two S_N^2 reactions |
| Threo - threo | |
| | Two Inversions $\equiv$ Retention. |

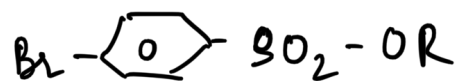


Symmetrical (trans)
(anti)

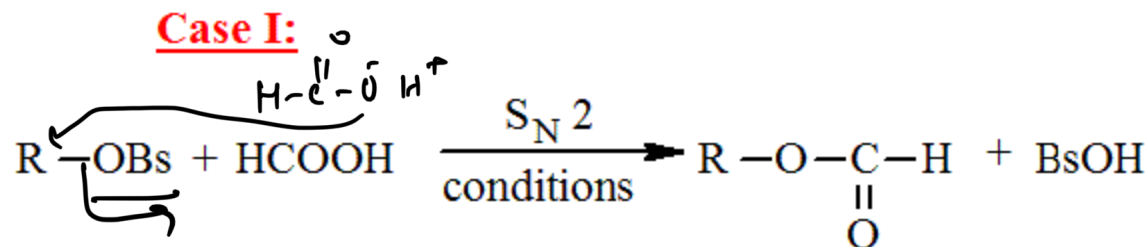


Retention of
config.

unsymmetrical

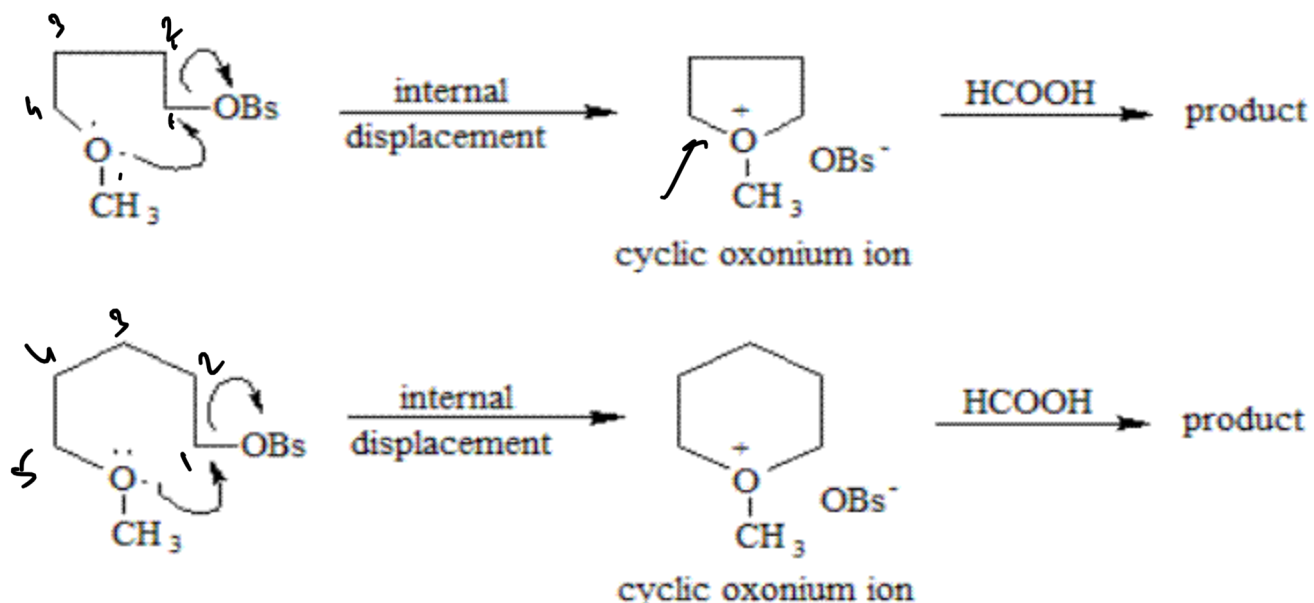


Consider the following data:



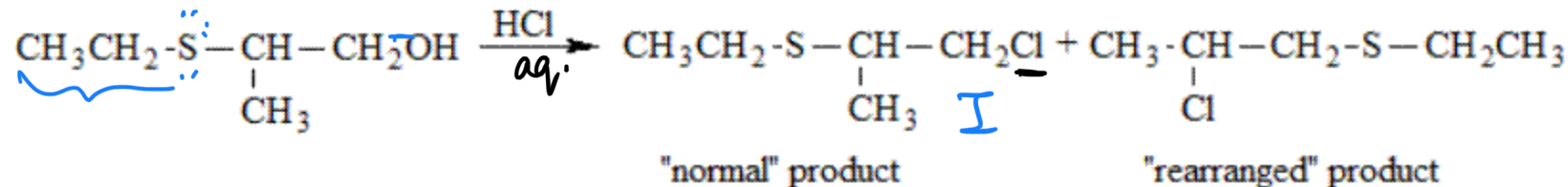
Compound #	R-OBs	k_{relative} (75°C)
1	$\text{CH}_3(\text{CH}_2)_2\text{CH}_2\text{-OBs}$	1.00 (reference)
2	$\text{CH}_3\text{OCH}_2\text{CH}_2\text{OBs}$	0.10
3	$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_2\text{OBs}$	0.33
4	$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OBs}$	461.0 ↓
5	$\text{CH}_3\text{O}(\text{CH}_2)_4\text{CH}_2\text{OBs}$	32.6 ↓
6	$\text{CH}_3\text{O}(\text{CH}_2)_5\text{CH}_2\text{OBs}$	1.13 ↓

Rationale to explain enhanced rates of substitution for compounds 4 & 5:

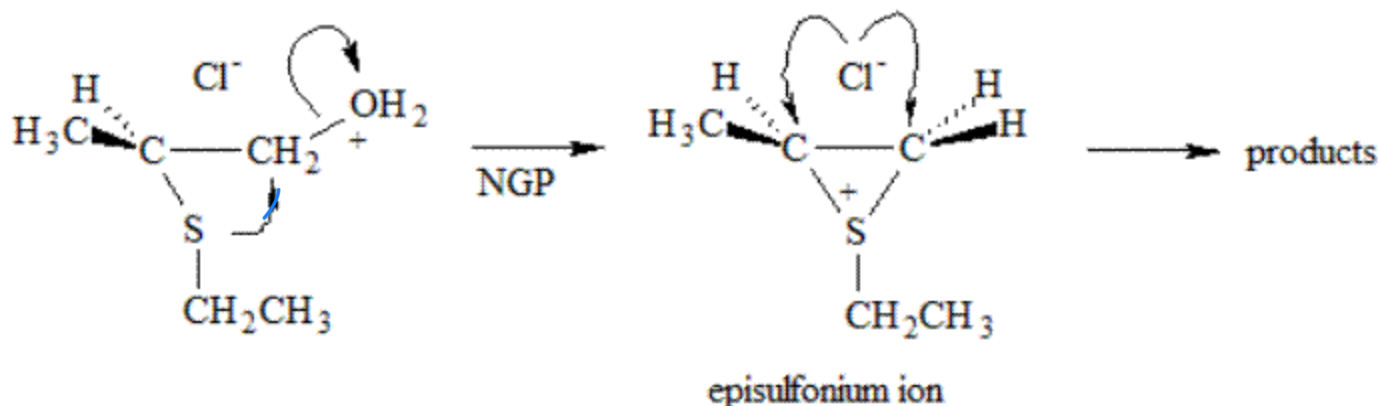


In addition to the available S_N2 pathway, a more favorable alternate pathway for displacement of brosylate is possible for compounds 4 & 5. The internal displacement pathway leads to the formation of relatively stable 5- and 6-membered cyclic oxonium salts. As the chain length increases, the likelihood of cyclic oxonium ion formation diminishes, and rates approach that of the reference case where only S_N2 attack by HCOOH is possible.

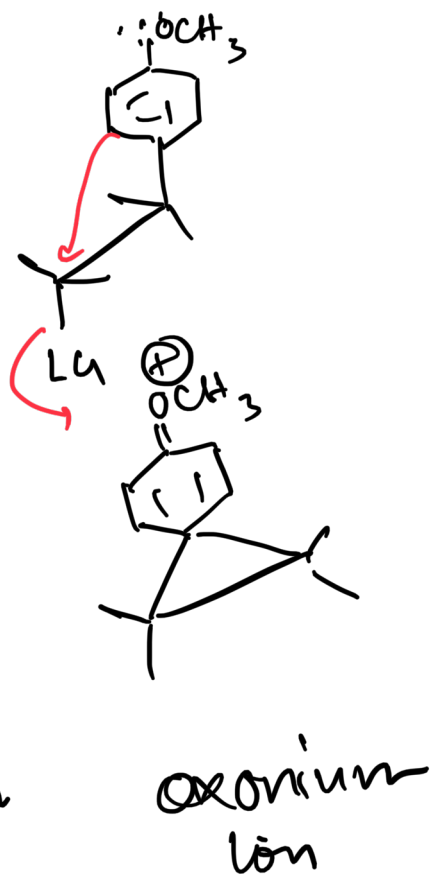
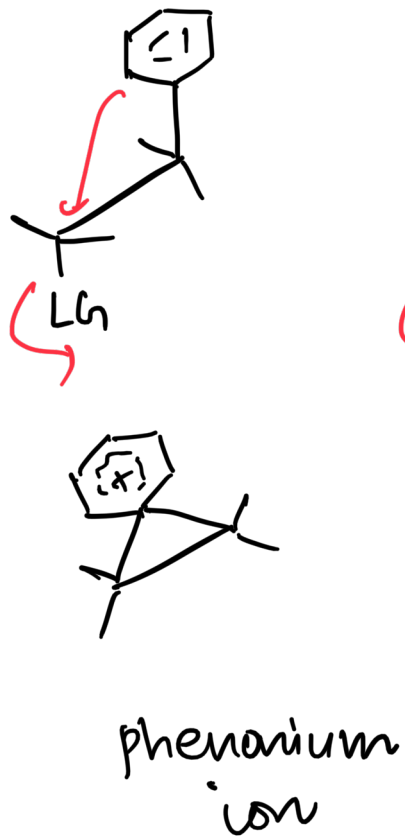
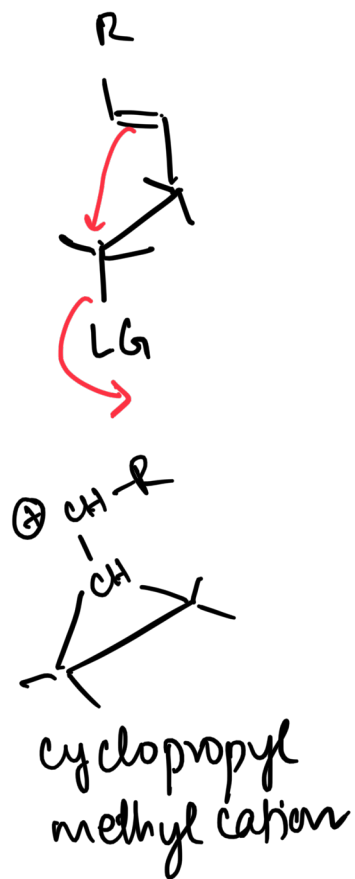
Consider the following:



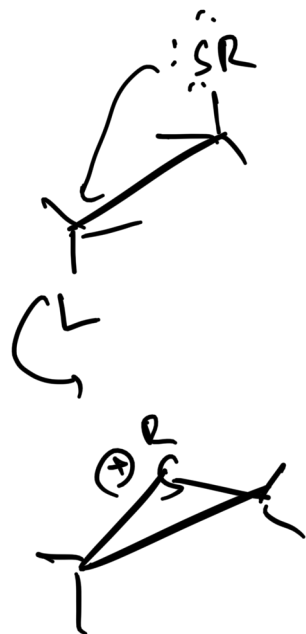
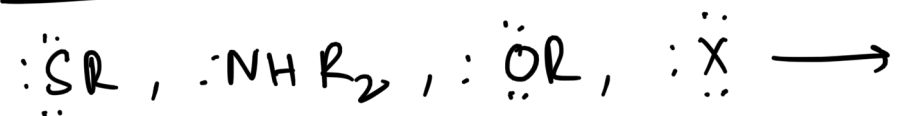
Rationale:



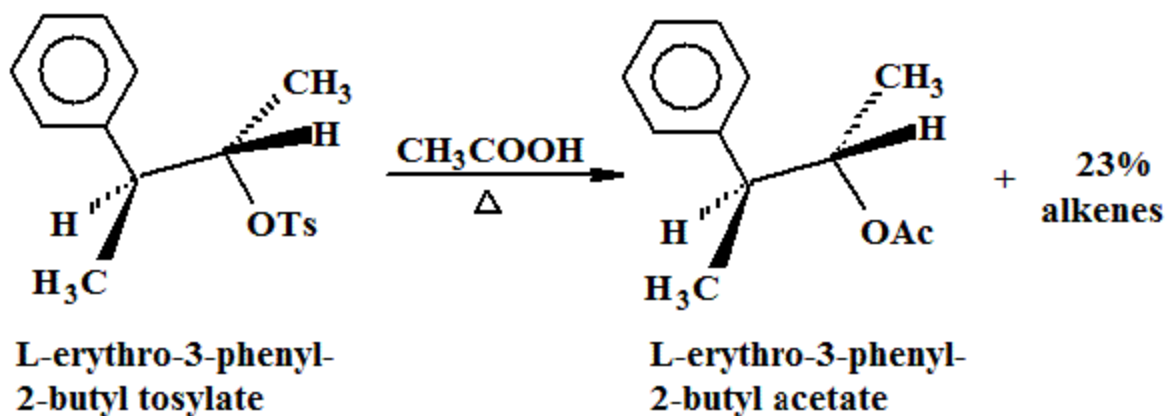
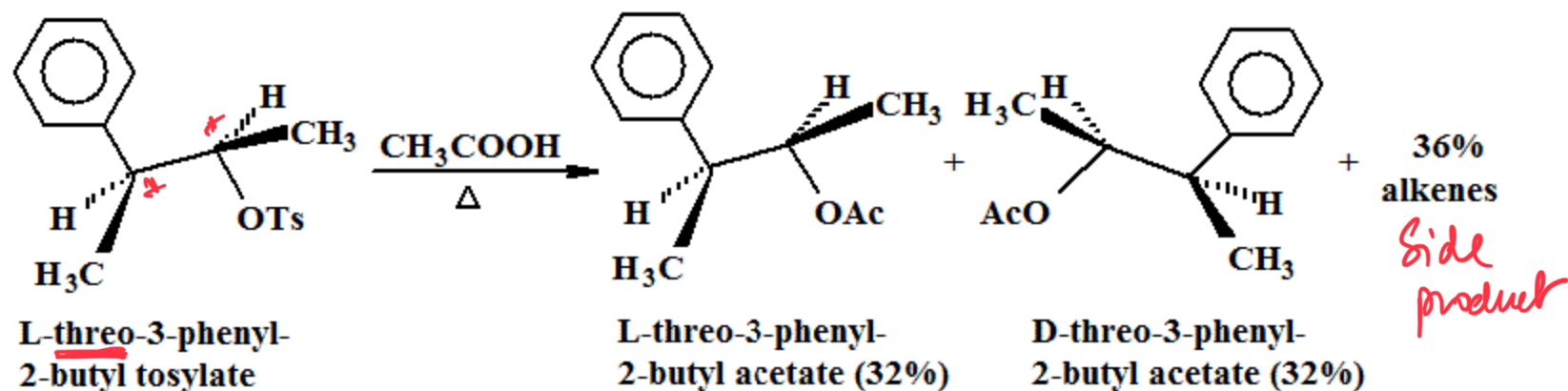
An $\text{S}_{\text{N}}1$ pathway leading to a primary carbocationic intermediate is not as favorable as a neighboring group participation (internal displacement) pathway leading to an episulfonium ion intermediate.

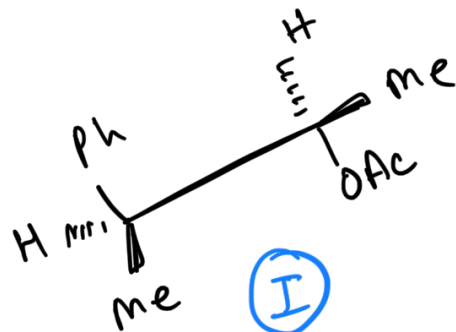
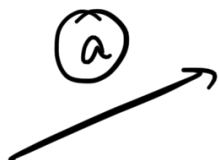
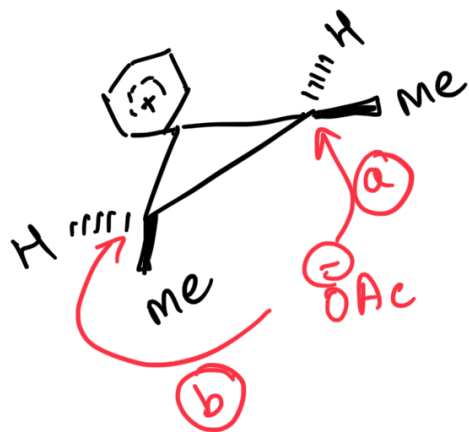


Mechanism of E_2

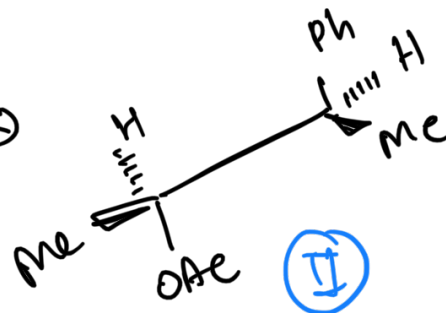
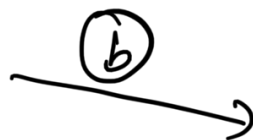


Consider the following:



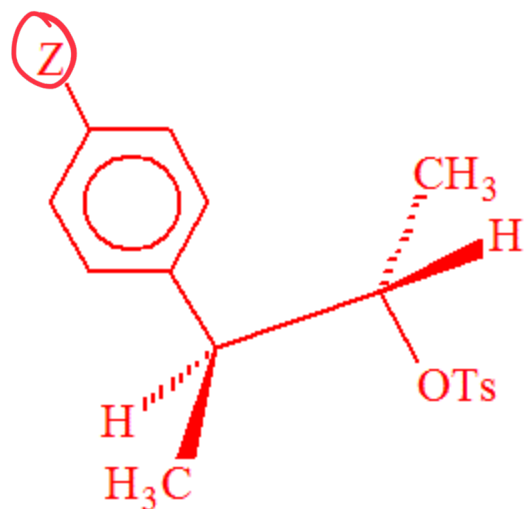


Ⓘ



Ⓣ

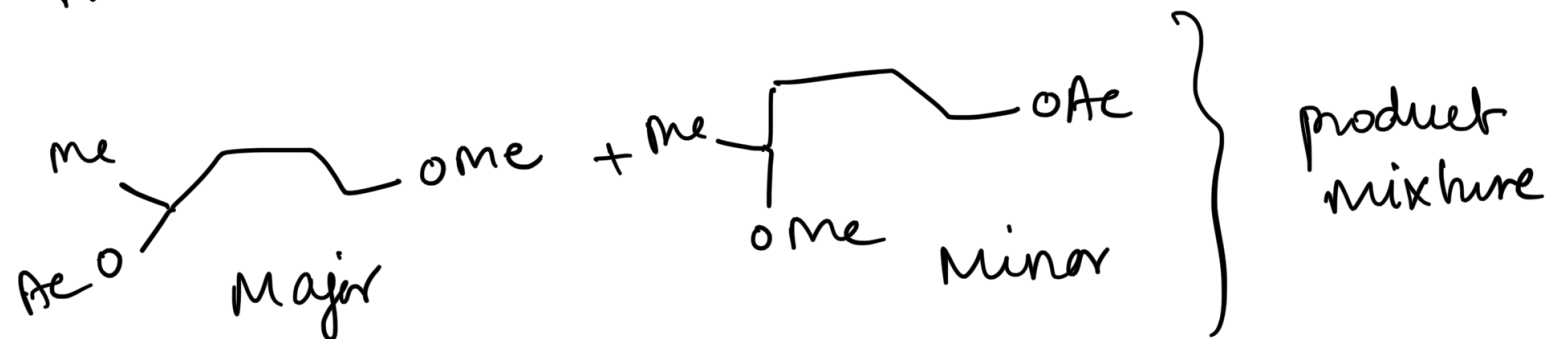
Both are
threo
Related
as
enantiomers.



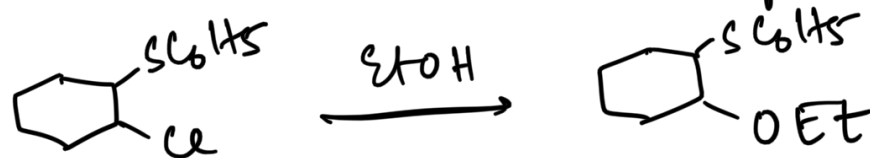
As the electron-donating ability of substituent Z increases, the rate of acetolysis increases.

	Cl	H	CH ₃	OCH ₃
$k_{\text{acetolysis}} (\times 10^7)$	0.39	2.39	19.0	228.0

Q: 4-methoxy-1-pentylbrosylate as well as 5-methoxy-2-pentylbrosylate on acetolysis give a mixture of the following two products. Explain the mechanism.



Q. Explain the rate of solvolysis for the reactions below.



trans : cis

70,000 : 0.16